METRIC

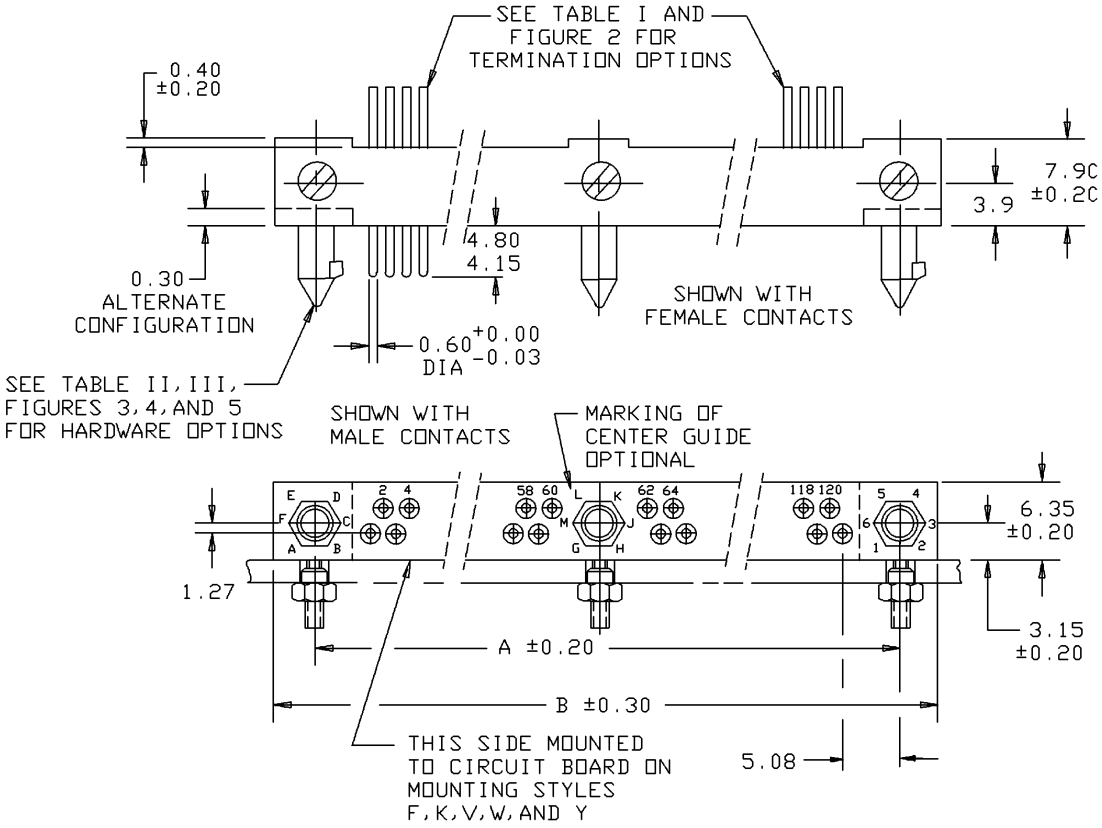
MIL-DTL-55302/163D
8 July 2004
SUPERSEDING
MIL-C-55302/163C(CR)
17 March 1995

DETAIL SPECIFICATION SHEET

CONNECTORS, PRINTED CIRCUIT SUBASSEMBLY AND ACCESSORIES, PLUG, 72, 84, 96, AND 120 CONTACT POSITIONS, FOR PRINTED WIRING BOARDS, 2.54 MM X 1.27 MM OFFSET GRID, METRIC

This specification is approved for used by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and MIL-DTL-55302.



NOTE: Dimensions are in millimeters.

FIGURE 1. Connector receptacle.

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RECEPTACLE TERMINAL STYLES

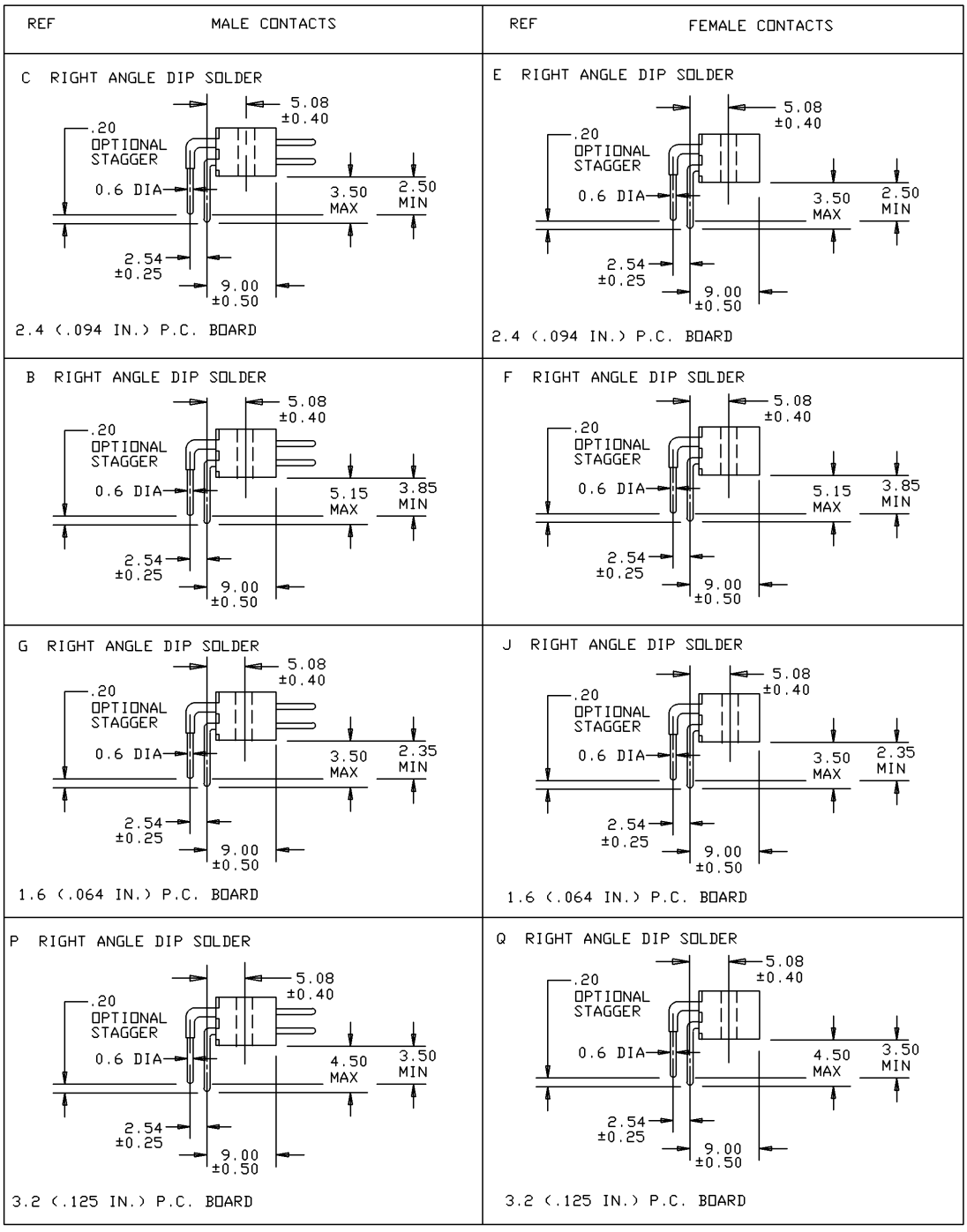


FIGURE 2. Connector termination styles.

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TERMINAL STYLES

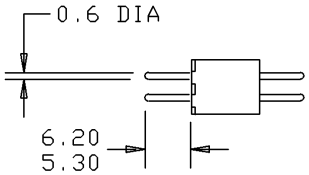
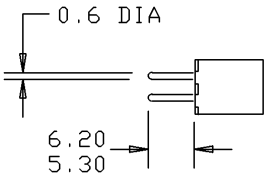
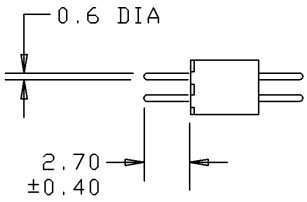
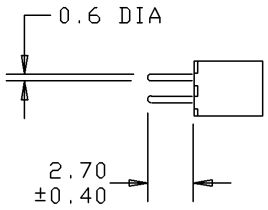
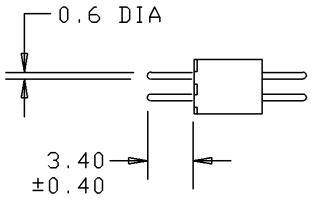
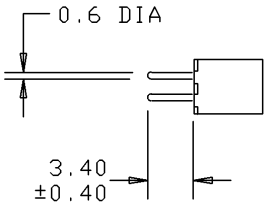
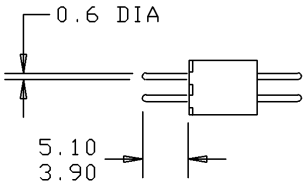
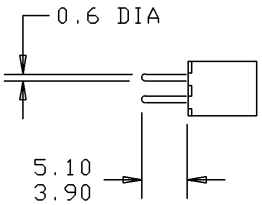
REF MALE CONTACTS	REF FEMALE CONTACTS
A STRAIGHT DIP SOLDER  	D STRAIGHT DIP SOLDER 
T STRAIGHT DIP SOLDER  1.6 (.064 IN.) P.C. BOARD N STRAIGHT DIP SOLDER	U STRAIGHT DIP SOLDER  1.6 (.064 IN.) P.C. BOARD Z STRAIGHT DIP SOLDER
 2.4 (.094 IN.) P.C. BOARD V STRAIGHT DIP SOLDER	 2.4 (.094 IN.) P.C. BOARD W STRAIGHT DIP SOLDER
 3.2 (.125 IN.) P.C. BOARD	 3.2 (.125 IN.) P.C. BOARD

FIGURE 2. Connector termination styles – Continued.

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TERMINAL STYLES

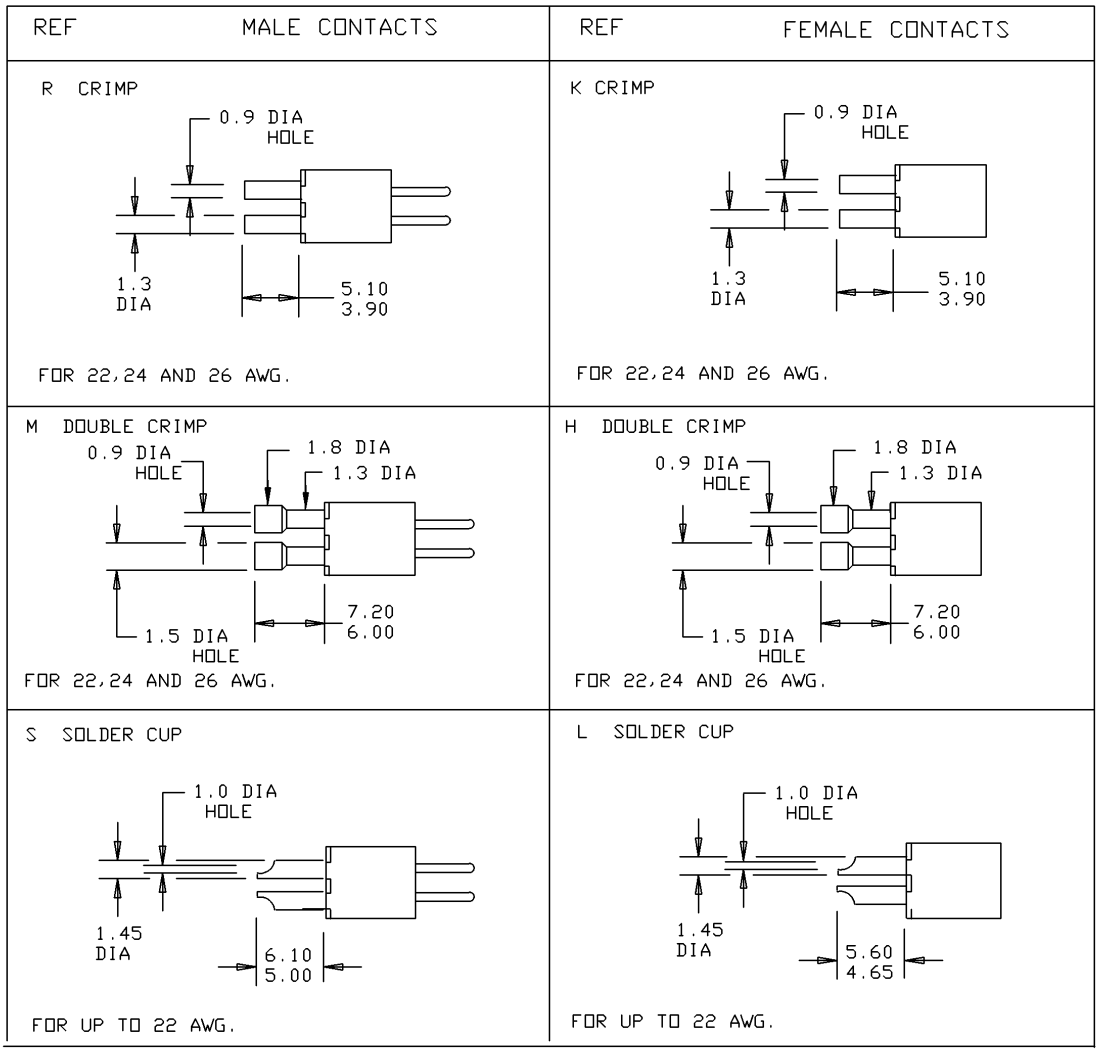
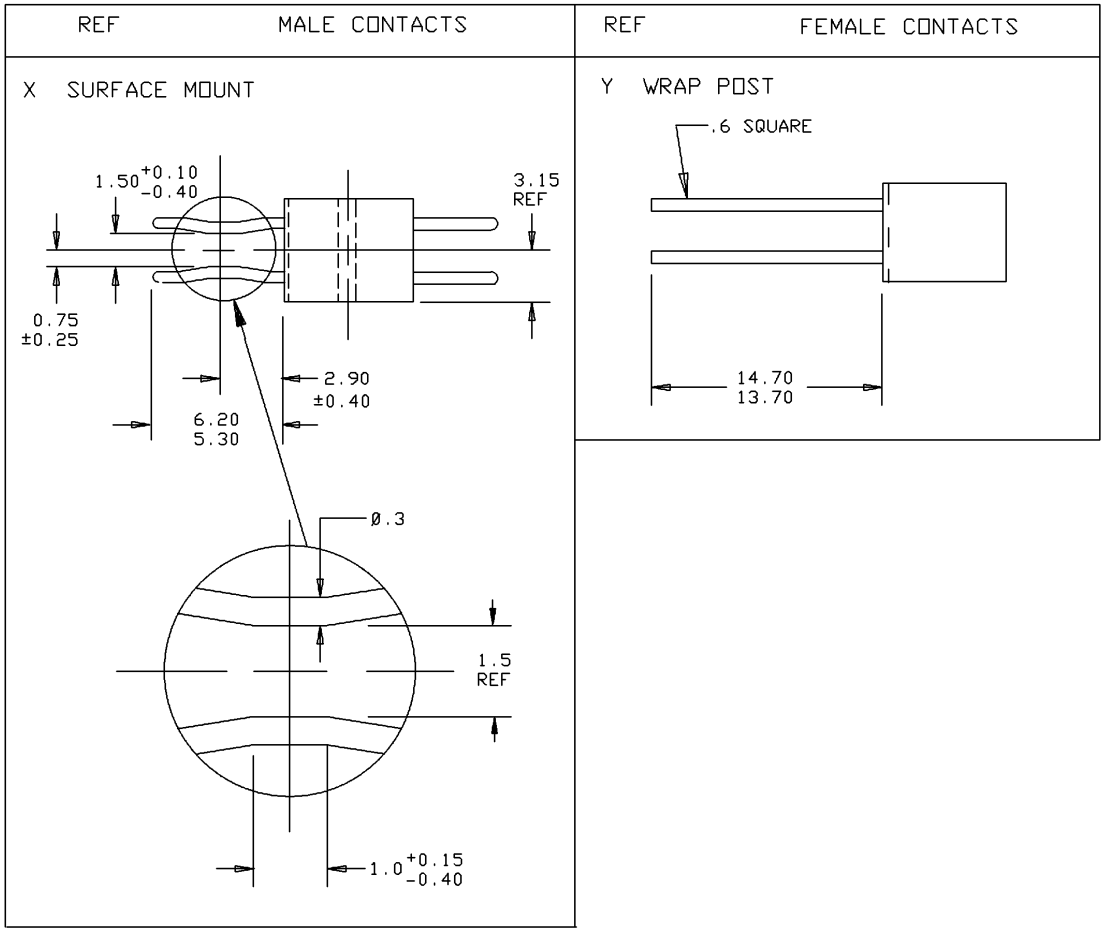


FIGURE 2. Connector termination styles – Continued.

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TERMINAL STYLES



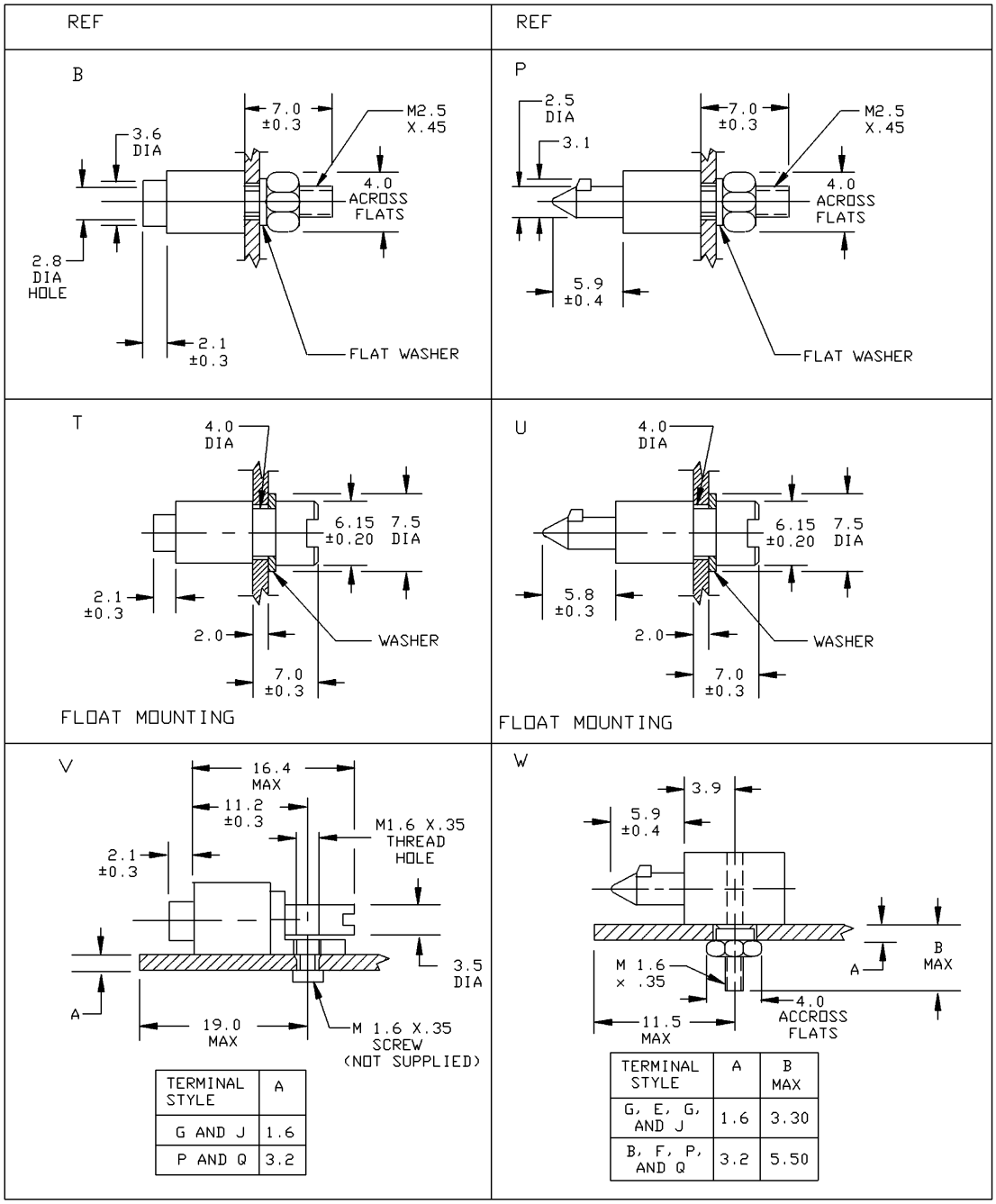
NOTE: Dimensions are in millimeters.

FIGURE 2. Connector termination styles – Continued.

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PLUG

MOUNTING STYLES



TERMINAL STYLE	A
G AND J	1.6
P AND Q	3.2

TERMINAL STYLE	A	B MAX
G, E, G, AND J	1.6	3.30
B, F, P, AND Q	3.2	5.50

NOTE: Dimensions are in millimeters.

FIGURE 3. Mounting styles.

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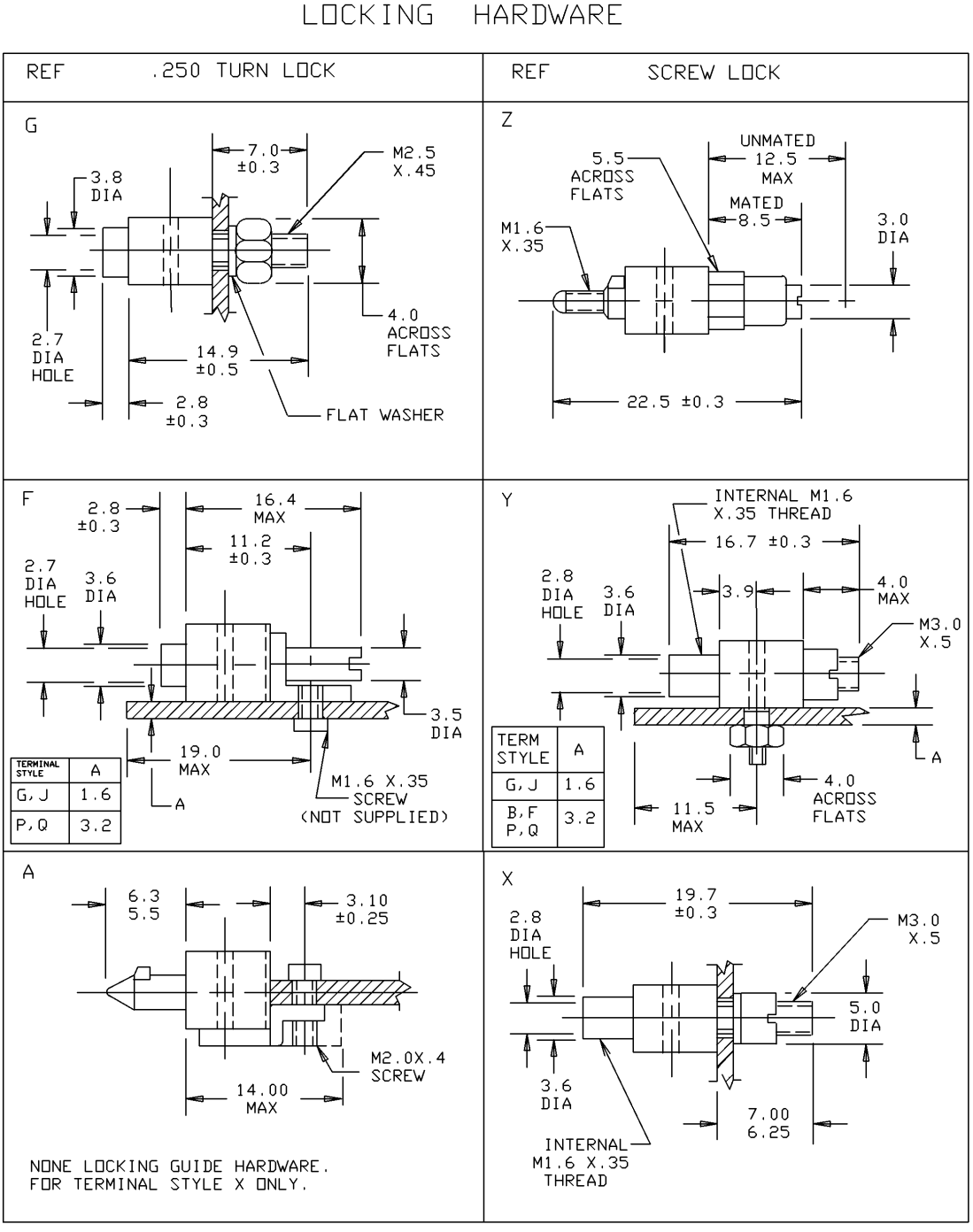
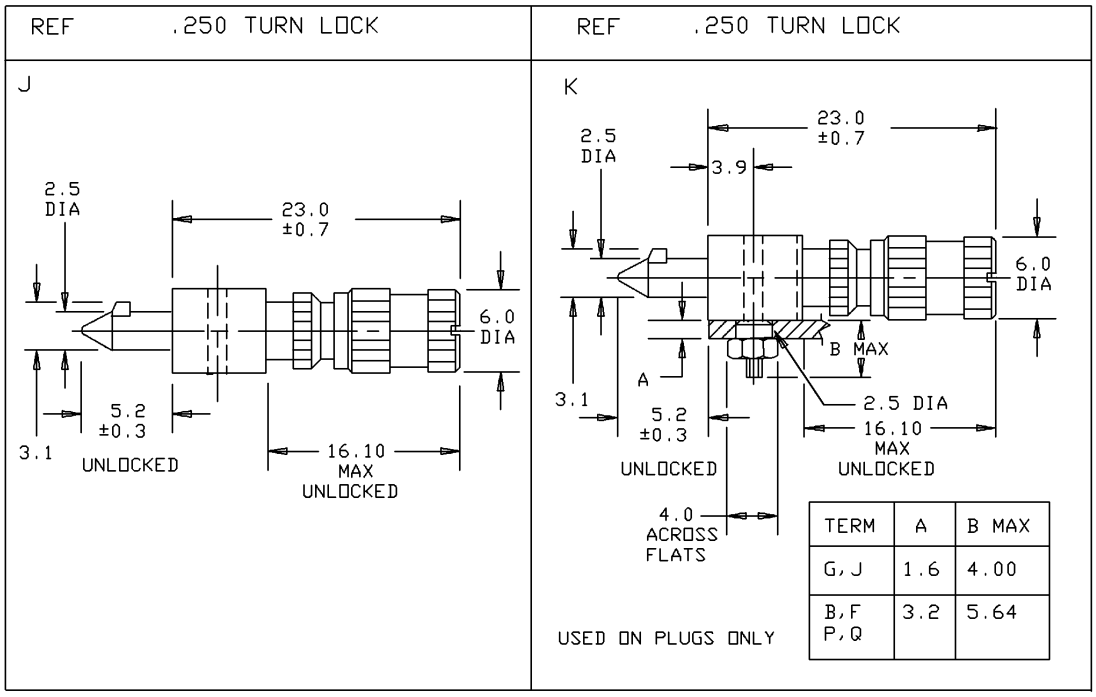


FIGURE 4. Locking hardware.

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LOCKING HARDWARE



TERM	A	B MAX
G, J	1.6	4.00
B, F P, Q	3.2	5.64

NOTES:
1. Dimensions are in millimeters.
2. This is a metric design.
3. Unless otherwise specified, tolerance is ± 0.10 millimeter.
4. These connectors mate with MIL-DTL-55302/160.
5. See table II for mating.
6. Contact identification may be located adjacent to cavity on face or side and may be rotated 180°.
7. Crimp contacts are shipped not inserted into insulator.
8. Use crimp tool M22520/2-01 for R and K terminal styles with Daniels manufacturing positioner K547.
9. Use crimp tool M22520/2-01 for M and H terminal styles with Daniels manufacturing positioner K547 and K640.

FIGURE 4. Locking hardware – Continued.

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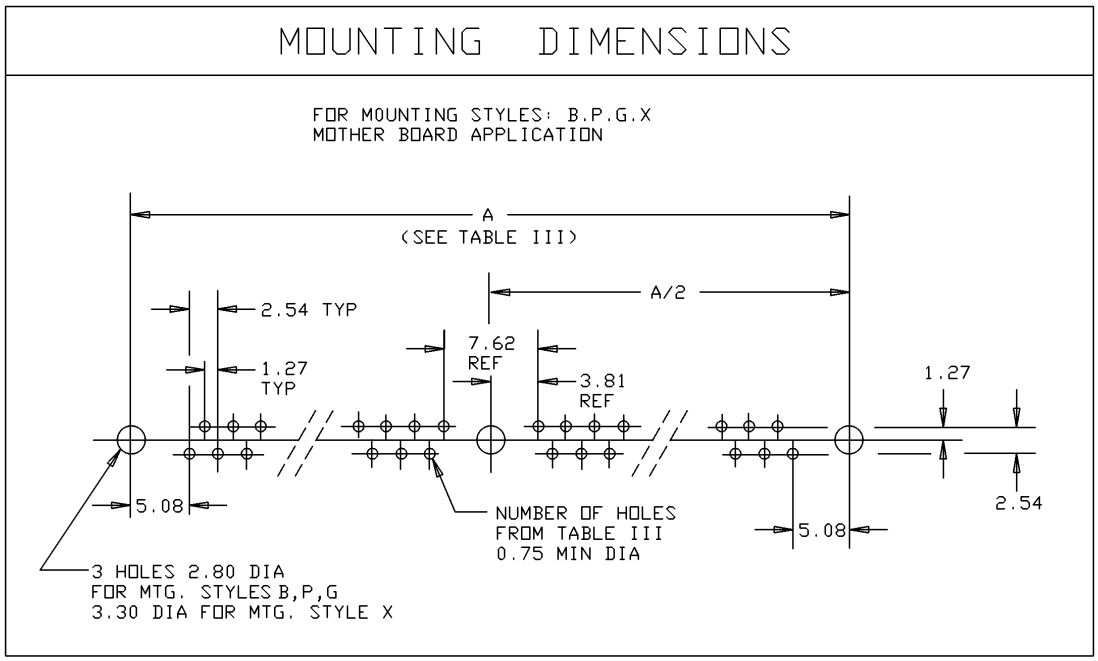


FIGURE 5. Mounting dimensions.

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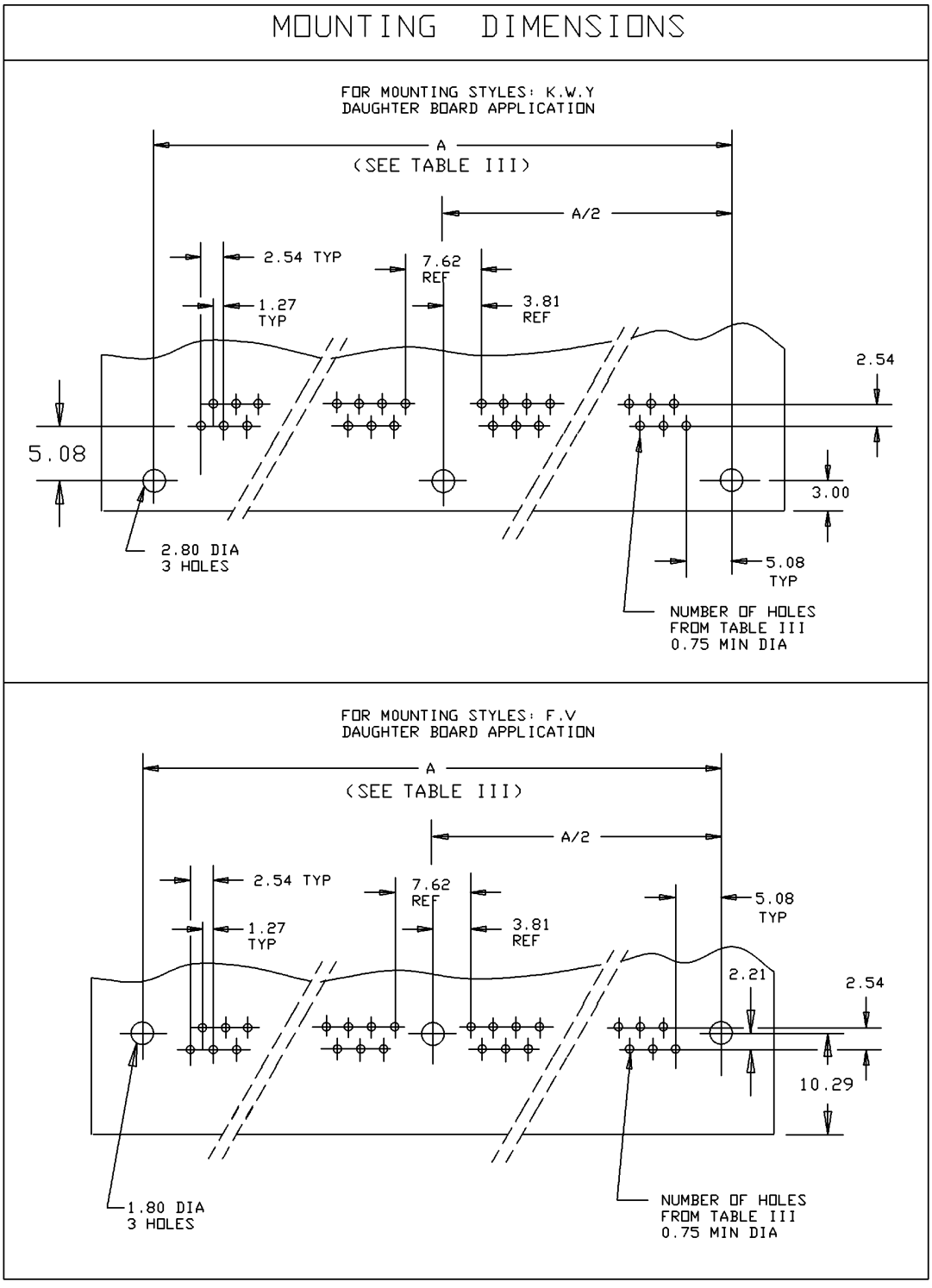
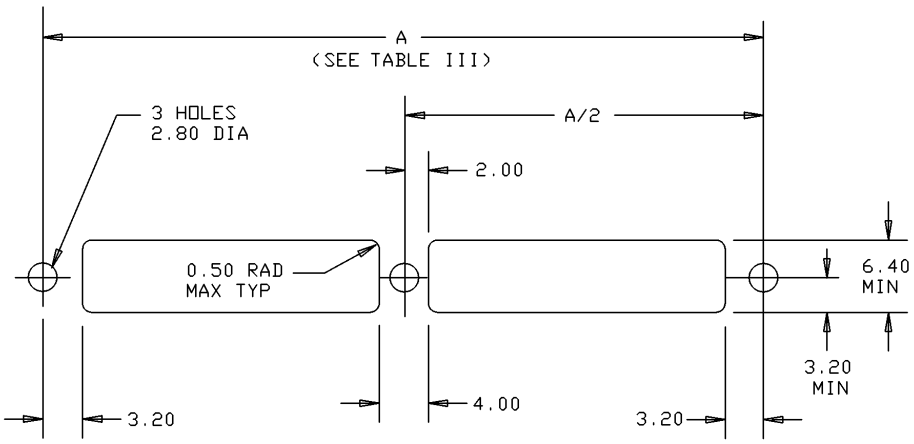


FIGURE 5. Mounting dimensions – Continued.

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MOUNTING DIMENSIONS

PANEL CUTOUT FOR FIXED
MOUNTING STYLES B,G,P



PANEL CUTOUT FOR FLOAT
MOUNTING STYLES T,U

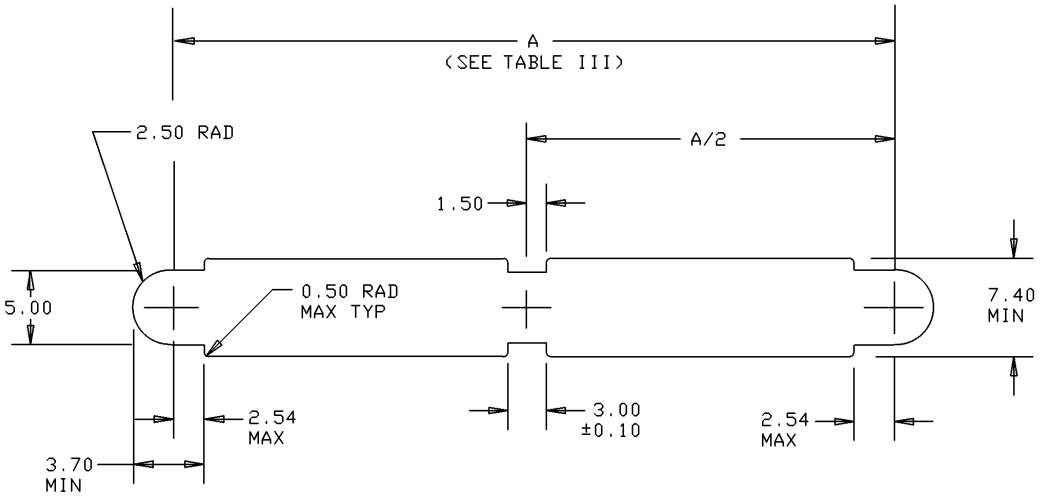
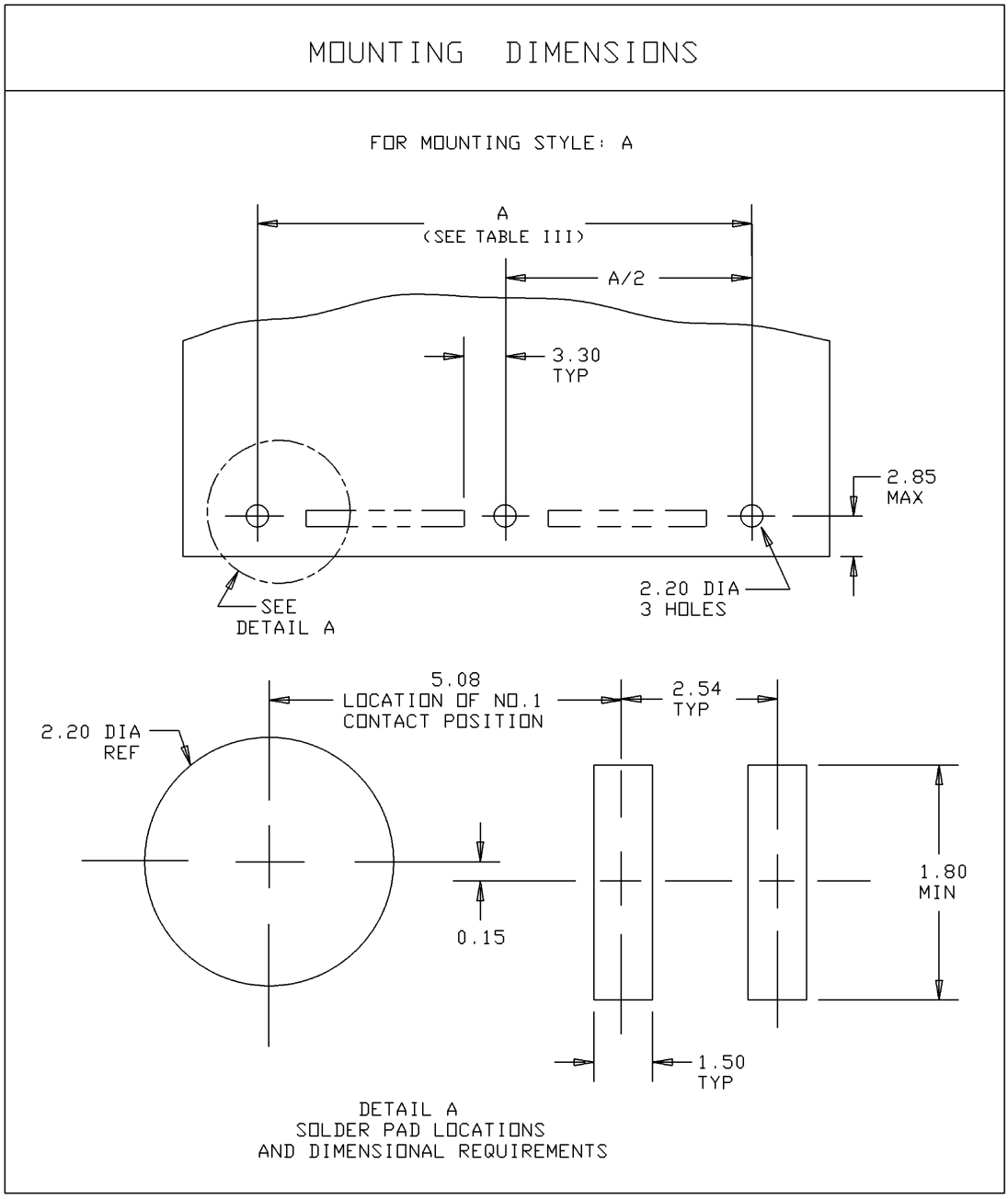


FIGURE 5. Mounting dimensions – Continued.

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NOTES:

1. For plugs only. The p.c. board may be extended to 19.00 max from center of mounting holes to serve as a pin protector.
2. Tolerance is noncumulative.
3. Dimensions are in millimeters.

FIGURE 5. Mounting dimensions – Continued.

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TABLE I. Number of contacts.

Number of contacts	Dimension (in millimeters)	
	A	B
72	106.7	114.7
84	121.9	129.9
96	137.2	145.2
120	167.6	175.5

TABLE II. Mating references.

Type	Will only mate with (see note)	Locking method
B	P, U, W	None
F	J, K	Push 1/4 turn
G	J, K	Push 1/4 turn
J	F, G	Push 1/4 turn
P	B, T, V	None
T	P, U, W	None
U	B, T, V	None
V	P, U, W	None
W	B, T, V	None
X	Z	Screw
Z	X, Y	Screw

NOTE: These connectors mate with MIL-DTL-55302/160.

TABLE III. Mounting references.

Number of contacts	A
72	106.70
84	121.90
96	137.20
120	167.60

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REQUIREMENTS

Design and construction:

Dimensions and configurations: See figures 1 through 5 and tables I, II, and III.

Material:

Insulators: Diallyl phthalate in accordance with ASTM D5948.

Guides and hardware: Brass in accordance with ASTM B36, B16, B124, B453, or B455; nickel plated in accordance with SAE-AMS-QQ-N-290 or AISI 300 series stainless steel or equivalent in accordance with ASTM A582, passivated in accordance with SAE-AMS-QQ-P-35.

Plating: Plating in accordance with MIL-DTL-55302 or as noted.

Mating surface (pin and wire): Gold in accordance with ASTM B488, class 1.27, type II, code C over nickel in accordance with SAE-AMS-QQ-N-290, 30-150 microinches.

Terminations: Solderable areas (including crimp barrels and wire wrap terminations) shall be gold in accordance with ASTM B488, type III, code A, class 0.254 minimum thickness or gold in accordance with ASTM B488, type II, code C, class 0.51 over nickel in accordance with SAE-AMS-QQ-N-290, 30 to 150 microinches thick. Solderable areas must meet the requirements of MIL-STD-202, method 208.

All other surfaces: Gold in accordance with ASTM B488, type II, code C, class 0.127 minimum over nickel in accordance with SAE-AMS-QQ-N-290, 30-150 microinches.

Contact identification: See figure 1 and table I.

Contact separation forces:

Minimum force: 0.14 newton (0.50 ounce) with a 0.5800 millimeter (.0228 inch) +0.005, -0.000 diameter pin after being cycled 3 times.

Note: Steel pin test surface roughness to be 0.025 – 0.25 micron rms.

Unmating: The maximum force of a connector assembly shall not exceed 0.53 newton (1.90 ounces) multiplied by the number of contact positions.

Mating: The maximum force of a connector assembly shall not exceed 0.70 newton (2.5 ounces) multiplied by the number of contact positions.

Contact retention minimum: Functional value 22.2 newtons (5 pounds); destructive test 35.3 newtons (8 pounds) maximum design and qualification value.

Contact rating: 5 amperes per line.

Durability: 2,000 cycles minimum.

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Dielectric withstanding voltage:

Sea level: 1,400 V ac rms test.
High altitude: 300 V ac rms test.

Keying: Rotation of guides, 6 positions each, 36 possible combinations.

Connector Part or Identifying Number (PIN):

Example: D55302/ 163 B 120 W

- Specification number
- Specification sheet number
- Type of contact and termination (see figures 1 and 2)
- Number of contacts (see table I)
- Guide hardware number (see figures 3, 4, 5 and tables II and III)

Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extent of the changes.

Referenced documents. In addition to MIL-DTL-55302, this document references the following:

MIL-DTL-55302/160	ASTM B36	ASTM B488
MIL-STD-202	ASTM B124	ASTM D5948
ASTM A582	ASTM B453	SAE-AMS-QQ-N-290
ASTM B16	ASTM B455	SAE-AMS-QQ-P-35

CONCLUDING MATERIAL

Custodians:
Army - CR
DLA - CC

Preparing activity:
DLA - CC

(Project 5935-4452-000)

Review activities:
Army - AR, AT, AV, CR4, MI

NOTE: The activities listed above were interested in this document as of the date of this document. Since organizations and responsible can change, you should verify the currency of the information above using the ASSIST Online database at http://www.dodssp.daps.mil.

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